Angel Mario Zarate Villazon

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Atlanta, Georgia

Education

- **Georgia Institute of Technology** — Atlanta, GA
  PhD in Aerospace Engineering (in progress) + MS in Aerospace Engineering; GPA: 4.0 — *August 2022 - May 2027 (expected)*
  - **Advanced Certificates**: Management of Technology Certificate for Graduate Students (Scheller College of Business)
  - **Honors**: NextProf Nexus program (co-organized by University of Michigan, Georgia Tech, UC Berkley): selected as one of 70 PhD candidates in engineering across the country; Daniel P. Schrage Award for Excellence: selected twice among 45+ other teams made up of ≈ 100 graduate students at the vehicle design Grand Challenge: one as a team leader and one as a team mentor.
- **Universidad Panamericana** — Mexico City, Mexico
  Bachelor of Mechanical Engineering; first in class, GPA: 9.89/10 — *August 2017 - June 2022*
  - **Advanced Certificates**: Specialty in robotics and manufacturing with diploma in data science
  - **Awards**: CENEVAL National Excellence Award (top $< 1\%$ national graduation exam) and ANFEI Best Engineering Graduate Award (single nomination per institution, national selection)

Research Experience

- **Aerospace Systems Design Laboratory, Georgia Tech** — Atlanta, GA
  Senior Graduate Researcher — *August 2022 - Present*

- **Lunar Surface Mobility Vehicle Fleet Architecting (dissertation)** — *November 2024 - Present*
  - **Vehicle family design for lunar rovers**: Recognized that family-based vehicle design processes from the automotive and aerospace industries could reduce duplicate work across the emerging family of lunar rover programs and better ensure stakeholder needs are met.
  - **Technology Enablers**: Built an AI agent-driven vehicle family design workflow that tests design alternatives from stakeholder analysis through simulation, ensuring consistency and traceability across system models to increase design efficiency and maximize shared technology investments across the fleet.
- **Mobility Digital Twin for FIFA World Cup 2026 in Atlanta** — *April 2025 - July 2026*
  - **Stakeholder Alignment**: Reconciled competing priorities across parking operations, on-the-ground police officers, the VP of Public Safety, and GDOT through stadium site visits and prototype-driven interviews to frame the core decision-support problem for FIFA World Cup 2026 mobility planning.
  - **Technical Development**: Determined the data and simulation capabilities needed to support the tool's decision framework, and coordinated with fellow ASDL researchers to integrate them; delivered an effective mobility digital twin for Downtown Atlanta to the Georgia World Congress Center Authority, Mercedes-Benz Stadium, and GDOT to cut decision turnaround from days to minutes.
- **Digital Twin Capability Research in Vehicle Dynamics** — *September 2022 - June 2025*
  - **Multi-Fidelity Vehicle Dynamics for Driver in the Loop Simulations (sponsored by Honda)**: Worked with Honda engineers to diagnose workflow inefficiencies in real-time driver-in-the-loop simulation, conceptualized a simulation testing workflow to identify drivers of error and computational cost; developed a tool with a team of fellow researchers that used machine learning to guide which modeling strategy to use; published at SAE World Congress 2026.
  - **High-fidelity Racing Car Simulation (sponsored by NASCAR OEM)**: Led a team to integrate a high-fidelity vehicle model into racing car simulations without sacrificing decision-making speed, using machine learning and sensitivity analysis, earning the Daniel P. Schrage Award for Excellence; returned the following year to focus on high-fidelity tire models as technical advisor for a repeat award-winning team; published at the American Modelica Conference and SAE World Congress 2026.
- **Sustainable Infrastructure and Electrification (sponsored by Siemens)** — *September 2022 - May 2023*
  - **Comprehensive Assessment of Electrification Solutions**: Worked with Siemens engineers and fellow ASDL researchers to determine the most cost-efficient renewable energy mix to meet growing electrification demands (EV charging, HVAC) in a corporate office building; presented the methodology and findings to Siemens Corporate Technology leadership, including the Head of Corporate Technologies; published at AIAA SciTech 2024.

Leadership Experience

- **International Council of the Aeronautical Sciences**
  Liaison for Mexico *September 2024 - Current*
  - **Building Representation**: Identified a lack of representation from Mexico in the council and brokered relationships between the Mexican Federation of the Aerospace Industry, the Mexican Council of Aerospace Education + 2 universities with ICAS leadership to establish the first delegation from Mexico, translating the ICAS value proposition into specific needs for the industry in Mexico.
  - **Academic Exchange**: Started academic exchange programs with two Mexican universities with Georgia Tech to support the growth of aeronautical research and development in Mexico that has led to admissions from PhD students and academic exchange weeks involving more than 20 students.
- **Aerospace Systems Design Laboratory, Georgia Tech**
  Head of the Academy *August 2025 - Current*
  - **Core Program Continuation**: Designed and moderated weekly peer-support sessions for 80+ researchers, setting discussion topics and facilitating technical debates, and escalating to Prof. Dimitri Mavris for expert intervention on complex methodological questions; drove a record number of internal proposals and earned recognition from the Vice Provost of Graduate Education at the Georgia Tech Graduate Education Expo.
  - **New Initiatives**: Identified gaps in logistical support for lab proposals and defenses and launched a volunteer program providing AV and catering coordination, improving event execution and strengthening cross-cohort collaboration.
- **Colibri Mission Aerospace Laboratory, Universidad Panamericana** Mexico City, Mexico
  Lead Systems Engineer *June 2018- July 2022*
  - **Leadership challenge**: Assembled and led a 50-person interdisciplinary team across two campuses to design Mexico's first 3-unit nanosatellite, organizing 10 technical disciplines to pass design reviews against a panel of experts from MIT, NASA, and the Mexican Space Agency; represented the technical effort in the fundraising campaign, including media engagement and executive-level negotiations.

Professional Experience

- **Toyota Racing Development** Salisbury, NC
  Vehicle Simulation Intern *May 2023 - August 2023*
  - **Vehicle Simulation Team**: Investigated novel optimization approaches for minimum lap-time simulation using high-fidelity vehicle models, under the mentorship of a former Formula 1 technical director; findings informed the team's approach to simulation-driven setup decisions.
- **Endodiagnostica S.C - Endoscopy Unit of Dalinde Hospital** Mexico City, Mexico
  Digital Transformation Consultant *July 2021 - July 2022*
  - **Digital Patient Experience**: Identified inefficiencies in the GI endoscopy patient experience and led a digital transformation project; selected and implemented a platform to deliver GI endoscopic studies results to patients digitally reducing result delivery costs by 30% after servicing more than 2000 patients in 6 months.
  - **Cross-functional collaboration**: Engaged cross-functional stakeholders (clinical, IT, administrative) and established requirements on the platform and its infrastructure to ensure organization-wide adoption.
  - **Staff Training**: Trained the team of 4 administrative assistants and 6 nurses on the processes related to the implementation of the digital environment.
- **Preparatoria Panamericana** Mexico City, Mexico
  F1 in Schools Workshop Coordinator *August 2017 - July 2022*
  - **Competition preparation**: Developed a competition-preparation program teaching engineering and business fundamentals to 100+ high school students; recruited and led co-instructors to scale delivery, and shortlisted 20 students for targeted mentoring. Teams qualified for 2 world finals, including a global Top 15 finish.

Core Strengths

- **Technical Foundation**: Physics-based simulation, digital twins, SysML v2, AI-driven decision workflows, multi-body dynamics, optimal control
- **Problem Structuring**: Stakeholder needs analysis, competing-priority reconciliation, requirements architecture, model-based systems engineering
- **Team Leadership**: Cross-functional research coordination, 80+ member peer network, 50-person interdisciplinary teams, sponsor engineering collaboration
- **Stakeholder Engagement**: Executive and C-suite briefings, cross-agency alignment, technical translation for non-technical audiences

Publications

Zarate Villazon, Angel M., Michael Balchanos, and Dimitri Mavris, "FEATURE MODEL CREATION THROUGH MISSION ANALYSIS IN THE CONTEXT OF VEHICLE FAMILIES", *ICAS*2026_0918, 2026

Zarate Villazon, Angel M., Erick Espinosa Juarez, Federico Perez Fuentes, and Dimitri Mavris, "TOWARDS AN ECOSYSTEM FOR AERONAUTICAL RESEARCH IN MEXICO", *ICAS*2026_0639, 2026 Balchanos, Michael, Mariam Emara, **Angel Zarate Villazon**, and Dimitri Mavris, "Integrating Optimization and Model Fidelity Selection Methods for Real-Time Driver-in-the-Loop Vehicle Simulators," SAE Technical Paper 2026-01-0621, 2026,

Zarate Villazon, Angel M., Ian Brown, Michael Balchanos, and Dimitri Mavris, "Lap Time Performance Sensitivity to Tire Model Parameters: An Optimal Control Framework for Model Selection and Calibration," SAE Technical Paper 2026-01-0214, 2026

Angel M. Zarate Villazon, Varick Peak, Scott Duncan, Olivia J. Pinon-Fischer and Dimitri N. Mavris. "A Multidisciplinary Approach to the Evaluation and Selection of Infrastructure Electrification Solutions," AIAA 2024-0855. AIAA SCITECH 2024 Forum. January 2024.

Seltikova, Ekaterina, et al. "A Framework for Mars Sample Return: Who Gets Access and What Issues Must Be Addressed." 35th IAA Symposium on Space Policy, Regulations and Economics (E3) The future of space exploration and innovation. 2022.

Zarate-Villazon, Ángel M., et al. "Mechanical development, production and testing of a miniaturized high fidelity accelerometer for a Cubesat." IAF Materials and Structures Symposium (C2) Specialized Technologies, Including Nanotechnology (8). 2022.

Zarate-Villazon, Ángel M., et al. "Integration of quality tools to define the scope of a Cubesat scientific/technology demonstration mission." IAF Space Systems Symposium (D1) Space Systems Engineering-Methods, Processes and Tools. 2021.